

Zuo Ye

Curriculum Vitae

Xipei Lu (Xinglong Duan), 266
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Working Experience

- 04/2025–present **Associate Professor** (a six-year position), *Institute of Mathematics and Interdisciplinary Sciences*, Xidian University, Xi'an, Shaanxi, China.
- 03/2022–03/2025 **Pos-Doc Fellow**, *School of Electrical and Computer Engineering*, Ben-Gurion University of the Negev, Beer Sheva, Israel.
Advisor: Ohad Elishco

Education

- 09/2018–11/2021 **Ph.D**, *School of Mathematical Sciences*, University of Science and Technology of China, Hefei, Anhui, China.
Advisor: Professor Xiande Zhang and Professor Gennian Ge
- 09/2016–06/2018 **Master of Science**, *School of Mathematical Sciences*, University of Science and Technology of China, Hefei, Anhui, China.
Advisor: Professor Xiande Zhang
- 09/2012–06/2016 **Bachelor of Science**, *School of Mathematical Sciences*, Tongji University, Shanghai, China.

Honors and Awards

- 2025 Xiaomi Young Scholar
- 2021 First-class Scholarship for doctoral students (University of Science and Technology of China)
- 2020 Second-class Scholarship for doctoral students (University of Science and Technology of China)
- 2019 First-class Scholarship for doctoral students (University of Science and Technology of China)
- 2015 China National Scholarship for undergraduate students
- 2015 First-class Scholarship for undergraduate students (Tongji University)
- 2015 Third Prize, National English Competition for College Students (NECCS) (Class C)
- 2014 Third Prize, the sixth Shanghai Undergraduate Mathematics Competition (Mathematics Major Group)

Research Interests

Applications of combinatorics and algebra in coding theory, coding for DNA-based data storage

Grants

2026.01– National Natural Science Foundation of China (Grant No. 12501466)
2028.12

Preprints

1. **Zuo Ye** and Gennian Ge, "Deletion-Correcting Codes for the ℓ -Symbol Read Channel", *submitted*.
2. **Zuo Ye**, Yuling Li, Zhaojun Lan and Gennian Ge, "Bounds and Constructions of Codes for Ordered Composite DNA Sequences", *submitted*.
3. **Zuo Ye** and Gennian Ge, "On the Maximum Size of Codes under the Damerau-Levenshtein Metric", *in preparation*.

Publications

1. **Zuo Ye**, Yubo Sun, and Gennian Ge, "Correcting Bursty/Localized Deletions: A New Error-Position-Estimation Code", *IEEE Transactions on Information Theory*, vol. 72, no. 3, pp. 1667–1682, 2026, doi: 10.1109/TIT.2026.3655098.
2. **Zuo Ye**, Yubo Sun, Wenjun Yu, Gennian Ge and Ohad Elishco, "Codes Correcting Two Bursts of Exactly b Deletions", *IEEE Transactions on Information Theory*, vol. 72, no. 4, pp. 1–1, 2026, doi: 10.1109/TIT.2025.3607876.
3. **Zuo Ye**, Omer Sabary, Ryan Gabrys, Eitan Yaakobi and Ohad Elishco, "More on Codes for Combinatorial Composite DNA", *Designs, Codes and Cryptography*, vol. 93, no. 5, pp. 3437–3462, 2025, doi: 10.1007/s10623-025-01634-8.
4. Wenjun Yu, **Zuo Ye** and Moshe Schwartz, "On the Asymptotic Rate of Optimal Codes that Correct Tandem Duplications for Nanopore Sequencing", *IEEE Transactions on Information Theory*, vol. 71, no. 5, pp. 3569–3581, 2025, doi: 10.1109/TIT.2025.3544875.
5. **Zuo Ye**, Wenjun Yu and Ohad Elishco, "Codes Over Absorption Channels", *IEEE Transactions on Information Theory*, vol. 70, no. 6, pp. 3981–4001, 2024, doi: 10.1109/TIT.2023.3346882.
6. **Zuo Ye** and Ohad Elishco, "Reconstruction of a Single String from a Part of its Composition Multiset", *IEEE Transactions on Information Theory*, vol. 70, no. 6, pp. 3922–3940, 2024, doi: 10.1109/TIT.2023.3315784.
7. **Zuo Ye**, Xin Liu, Xiande Zhang, and Gennian Ge, "Reconstruction of Sequences Distorted by Two Insertions", *IEEE Transactions on Information Theory*, vol. 69, no. 8, pp. 4977–4992, 2023, doi: 10.1109/TIT.2023.3260872.
8. **Zuo Ye**, Tao Zhang, Xiande Zhang and Gennian Ge, "Some New Results on Splitter Sets," *IEEE Transactions on Information Theory*, vol. 66, no. 5, pp. 2765–2776, May 2020, doi: 10.1109/TIT.2019.2952370.

9. Yu Ning, **Zuo Ye**, Gennian Ge, Fuyou Miao, Yan Xiong, and Xiande Zhang, "New Results on Self-Dual Generalized Reed-Solomon Codes", *IEEE Transactions on Information Theory*, vol. 67, no. 11, pp. 7240–7252, Nov. 2021, doi: 10.1109/TIT.2021.3096934.
10. Fei Shi, **Zuo Ye**, Lin Chen, and Xiande Zhang "Strong quantum nonlocality in any N-partite system", *Physical Review A*, vol. 105 no. 2, Feb. 2022, doi: 10.1103/PhysRevA.105.022209

Talks

1. Workshop on Combinatorics and Coding Theory, Kunming, China, July 5–11, 2026
2. "Constructions of Codes Correcting Burst-Deletions and Localized Deletions", [The 12th Korea-Sino/Sino-Korea International Conference on Coding Theory and Related Topics](#), Kangwon National University in Chuncheon, Korea, July 19–23, 2026, **an invited talk, 40 minutes**
3. "Codes Correcting Two Bursts of Exactly b Deletions", [The Ninth Symposium on Coding Theory, Cryptography and Combinatorics](#), Qingdao, China, May 8–12, 2025.
4. "Polymer-Based Storage and Reconstruction of Strings from Their Prefix-Suffix Compositions", Youth Forum on Extremal Combinatorics and Information Related Problems, Soochow University, China, October 22, 2022, **an invited online talk, 50 minutes**
5. "Some New Results on Splitter Sets", [The 10th International Symposium on Finite Fields and Their Applications](#), Luoyang, China, May 20–24, 2021.

Teaching Experience

1. **Combinatorics**, Teaching Assistant, University of Science and Technology of China, 2017 Fall.
2. **Algebraic Number Theory**, Co-instructor, Capital Normal University, 2021 Fall.
3. **Linear Algebra**, Teaching Assistant, Xidian University, 2025 Fall.
4. **Mathematical Statistics**, Xidian University, 2026, Fall
5. **Mathematics in Information**, Xidian University, 2026, Fall

Journal/Conference paper Referee

1. IEEE Transactions on Information Theory
2. IEEE Transactions on Molecular, Biological, and Multi-Scale Communications
3. Designs, Codes and Cryptography
4. IEEE Journal on Selected Areas in Information Theory
5. IEEE International Symposium on Information Theory (ISIT): 2024, 2025
6. 2024 IEEE Information Theory Workshop (ITW)